

spectra (i.e., the formation of spectral lines in the case of atoms), and the decrease of the specific heat of solids at low temperatures were explained clearly.

Example 8.1 : Calculate the energy in eV for a photon of wavelength $0.1 \times 10^{-9} \text{ m}$. What is the momentum of this photon.

Solution:

Given, $\lambda = 0.1 \times 10^{-9} \text{ m}$; $E = ?$; $p = ?$

Formula

$$E = h\nu = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{0.1 \times 10^{-9}}$$

$$\therefore E = 1.986 \times 10^{-15} \text{ J.}$$

$$\text{or } E = \frac{1.986 \times 10^{-15}}{1.6 \times 10^{-19}} = 12412.5 \text{ eV.}$$

Momentum of the photon is

$$\therefore p = \frac{h}{\lambda} = \frac{6.62 \times 10^{-34}}{0.1 \times 10^{-9}}$$

$$p = 6.62 \times 10^{-24} \text{ kgms}^{-1}.$$

Example 8.2 : Calculate the number of photons emitted by a 100 watt sodium lamp. The wavelength of emitted light is $5893\text{\AA}$.

Solution :

Given, $\lambda = 5893\text{\AA} = 5893 \times 10^{-10}\text{m}$

Total energy emitted per sec = 100

Formula

The energy of one photon is

$$E = h\nu = \frac{hc}{\lambda} = \frac{6.62 \times 10^{-34} \times 3 \times 10^8}{5893 \times 10^{-10}}$$

$$E = 3.370 \times 10^{-19}\text{J}.$$

$\therefore$ The total number of photons emitted per second is

$$N = \frac{\text{Total energy emitted per sec}}{\text{energy of one photon}}$$

$$N = \frac{100}{3.370 \times 10^{-19}} = 2.967 \times 10^{20} \text{ per sec.}$$

Example 8.3 : Calculate the energy density per unit wavelength in a black body cavity at a temperature of 1500K at a wavelength of $4000\text{\AA}$.

Solution:

Given, $\lambda = 4000\text{\AA} = 4000 \times 10^{-10}\text{m}$; $T = 1500\text{K}$; $E_\lambda d\lambda = ?$

Formula

$$E_\lambda d\lambda = \frac{8\pi hc}{\lambda^5} \left[\frac{1}{e^{\left(\frac{hc}{\lambda k_B T}\right)} - 1} \right] \text{ For a unit wavelength range.}$$

$$\frac{hc}{\lambda k_B T} = \frac{6.625 \times 10^{-34} \times 3 \times 10^8}{4000 \times 10^{-10} \times 1.38 \times 10^{-23} \times 1500} = 24$$

$$\therefore e^{\left(\frac{hc}{\lambda k_B T}\right)} = e^{24} = 2.6585 \times 10^{10}$$

$$\text{Hence, } e^{\left(\frac{hc}{\lambda k_B T}\right)} - 1 = 2.6585 \times 10^{10} - 1 \\ = 2.6585 \times 10^{10}$$

$$\therefore \frac{1}{e^{\left(\frac{hc}{\lambda k_B T}\right)} - 1} = \frac{1}{2.6585 \times 10^{10}} = 3.761 \times 10^{-11}$$

Hence,
$$E_{\lambda} d\lambda = \frac{8 \times 3.14 \times 6.625 \times 10^{-34} \times 3 \times 10^8}{(4000 \times 10^{-10})^5} \times 3.761 \times 10^{-11}$$

$$E_{\lambda} d\lambda = 0.01833 \text{ J/m}^4.$$

- No quantum number can be zero. Because, if any one of them is zero, it means that the particle does not exist in the box, which is not physically acceptable.

3 **Example 8.9 :** Find the energy of an electron moving in one dimension in an infinitely high potential box of width 0.1 nm.

Solution :

Given $a = 0.1 \text{ nm} = 0.1 \times 10^{-9} \text{ m}$.

The energy of an electron is

$$E = \frac{n^2 h^2}{8ma^2} \quad (\text{where } n = 1, 2, 3, \dots)$$

The least energy of the particle can be obtained by taking $n = 1$. Thus,

$$E = \frac{h^2}{8ma^2} = \frac{(6.625 \times 10^{-34})^2}{9 \times 9.11 \times 10^{-31} \times (0.1 \times 10^{-9})^2}$$

$$= 6.022 \times 10^{-18} \text{ J} = \frac{6.022 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

or $E = 37.69 \text{ eV}$.

4 **Example 8.10 :** What is the lowest energy that a neutron can have if confined to move along the edge of an impenetrable box of length 10^{-14} m ? (Given mass of neutron = $1.67 \times 10^{-27} \text{ kg}$).

Solution :

Given, $a = 10^{-14} \text{ m}$; $m = 1.67 \times 10^{-27} \text{ kg}$.

The quantized energies of a particle is given by

$$E = \frac{n^2 h^2}{8ma^2} \quad (\text{where } n = 1, 2, 3, \dots)$$

For the lowest energy, $n = 1$

$$\therefore E = \frac{h^2}{8ma^2}$$

$$= \frac{(6.625 \times 10^{-34})^2}{8 \times 1.67 \times 10^{-27} \times (10^{-14})^2} = \frac{4.38906 \times 10^{-67}}{1.336 \times 10^{-54}}$$

$$= 2.053 \times 10^6 \text{ eV}$$

$$E = 2.053 \text{ MeV.}$$

Example 8.11 : Compute the first 3 permitted energy values for an electron in a box of width $4 \text{ \AA}$

Solution :

Given, $a = 4 \text{ \AA} = 4 \times 10^{-10} \text{ m}$; $E_0 = ?$; $E_1 = ?$; $E_2 = ?$

The energy of electron is

$$E_n = \frac{n^2 h^2}{8ma^2} \quad (\text{where } n = 1, 2, 3, \dots) \quad \text{---- [1]}$$

The first permitted energy level is obtained by taking $n = 1$ in equation (1)

$$\begin{aligned} \therefore E_1 &= \frac{h^2}{8ma^2} \\ &= \frac{(6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (4 \times 10^{-10})^2} = 3.7680 \times 10^{-19} \text{ J} \end{aligned}$$

$$\text{or } E_1 = \frac{3.7680 \times 10^{-19}}{1.6 \times 10^{-19}} \text{ eV}$$

$$\therefore E_1 = 2.355 \text{ eV.}$$

The second permitted energy level is obtained by taking $n = 2$.

$$\therefore E_2 = \frac{(2)^2 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (4 \times 10^{-10})^2}$$

$$\begin{aligned} \text{or } E_2 &= 4 E_1 \\ &= 4 \times 2.355 \text{ eV} \end{aligned}$$

$$E_2 = 9.420 \text{ eV.}$$

The third permitted energy level is obtained by taking $n = 3$

$$\begin{aligned} \therefore E_3 &= 9 E_1 \\ E_3 &= 9 \times 2.355 \text{ eV} \\ E_3 &= 21.195 \text{ eV.} \end{aligned}$$

Example 8.12 : Find the lowest energy of an electron confined in a cubical box of each side $1.5 \text{ \AA}$.

Solution :

Given, $a = 1.5 \times 10^{-10} \text{ m}$.

The possible energies of a particle in a cubical box of each side a are given by

$$E = \frac{n^2 h^2}{8ma^2} = \frac{h^2}{8ma^2} (n_x^2 + n_y^2 + n_z^2)$$

For the lowest energy $n_x = n_y = n_z = 1$

$$\therefore E = \frac{h^2}{8ma^2} \times 3$$

$$= \frac{3 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (1.5 \times 10^{-10})^2}$$

$$= 8.029 \times 10^{-18} \text{ J}$$

$$\text{or } E = \frac{8.029 \times 10^{-18}}{1.6 \times 10^{-19}} \text{ eV}$$

$$E = 50.18 \text{ eV.}$$

Example 8.13 : Calculate the minimum energy an electron can possess in an infinitely deep potential well of width 4 nm .

Solution :

Given, $a = 4 \text{ nm} = 4 \times 10^{-9} \text{ m}$; $E = ?$.

Formula

$$E = \frac{n^2 h^2}{8ma^2} \quad (\text{For minimum energy } n = 1)$$

$$\therefore E = \frac{1^2 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (4 \times 10^{-9})^2}$$

$$E = 3.7639 \times 10^{-21} \text{ J}$$

$$\text{or } E = 0.02352 \text{ eV.}$$

Example 8.14 : An electron is trapped in an one dimensional box of length 0.1nm. Calculate the energy required to excite the electron from its ground state to the fifth excited state.

Solution :

Given, $a = 0.1 \text{ nm} = 0.1 \times 10^{-9} \text{ m}$.

Formula
$$E = \frac{n^2 h^2}{8ma^2}$$

For ground state $n = 1$

$$\therefore E_G = \frac{1^2 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (0.1 \times 10^{-9})^2}$$

or $E_G = 6.022 \times 10^{-18} \text{ J}$.

For, fifth excited state $n = 6$

$$\therefore E_e = \frac{6^2 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (0.1 \times 10^{-9})^2}$$

$$E_e = 2.16803 \times 10^{-16} \text{ J}$$

$\therefore$ The energy required to excite the electron from its ground state to the fifth excited state is

$$E = E_e - E_G$$

Hence, $E = 1317 \text{ eV}$.

Example 8.15 : Calculate the lowest energy of the system consisting of three electron in a one dimension potential box of length 1 nm.

Solution :

Given, $a = 1 \text{ nm} = 1 \times 10^{-9} \text{ m}$. The number of electrons given is 3 ; $E = ?$

Formula

$$E = \left[\frac{n^2 h^2}{8ma^2} \right] \times 3$$

For ground state $n = 1$

$$\therefore E_G = \frac{1^2 \times (6.625 \times 10^{-34})^2}{8 \times 9.11 \times 10^{-31} \times (1 \times 10^{-9})^2} \times 3$$

or $E_G = 6.022 \times 10^{-18} \text{ J} \times 3$
 $= 1.8066 \times 10^{-17} \text{ J}$

$$\therefore E_G = 112.9125 \text{ eV}$$

microscope.

4.13 Solved Examples

Example 4.13.1 Calculate the De-Broglie wavelength of a 20 keV neutron. Given mass of neutron, $m = 1.67 \times 10^{-27}$ kg.

Solution :

$$E = 20 \text{ keV} \times 20 \times 10^3 \text{ eV}$$

$$m = 1.67 \times 10^{-27} \text{ kg}$$

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$$h = 6.63 \times 10^{-34} \text{ Js}$$

$$\lambda = \frac{h}{\sqrt{2 m E}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 1.67 \times 10^{-27} \times 20 \times 10^3 \times 1.6 \times 10^{-19}}}$$

$$\lambda = 2.02 \times 10^{-13} \text{ m}$$

Example 4.13.2 Calculate the wavelength of a photon and an electron both having an energy 2.0 eV.

Solution :

$$h = 6.63 \times 10^{-34} \text{ J-s, Mass of electron} = 9.1 \times 10^{-31} \text{ kg}$$

i) For electron, $\lambda = \frac{h}{\sqrt{2 m E}}$, ii) For photon,

$$E = h\nu = \frac{hc}{\lambda}$$

$$\therefore \lambda = \frac{hc}{E}$$

i) for electron,

$$\begin{aligned} \lambda &= \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.1 \times 10^{-31} \times 2 \times 1.6 \times 10^{-19}}} \\ &= 0.8687 \times 10^{-9} \text{ m} \\ &= 8.687 \text{ \AA} \end{aligned}$$

ii) For photon,

$$\begin{aligned} \lambda &= \frac{hc}{E} \\ &= \frac{6.63 \times 10^{-34} \times 3 \times 10^8}{2 \times 1.6 \times 10^{-19}} \\ &= 6.2156 \times 10^{-7} \text{ m} \\ &= 6215.6 \text{ \AA} \end{aligned}$$

Example 4.13.3 What potential difference must be applied to an electron microscope to obtain electrons of wavelength 0.6 \text{ \AA}.

Solution : $\lambda = 0.6 \text{ \AA}$

We know, for electron, $\lambda = \frac{12.27}{\sqrt{V}} \text{ \AA}$

$$\therefore \sqrt{V} = \frac{12.27}{\lambda}$$

$$\therefore \sqrt{V} = \frac{12.27}{0.6}$$

$$V = 418.2025 \text{ volt}$$

Example 4.13.4 Compare the lowest three energy states for an electron confined in an infinite potential well of width $10\text{\AA}$.

Solution : $L = 10\text{\AA} = 10 \times 10^{-10} \text{ m}$

$$\begin{aligned} \therefore E_n &= \frac{n^2 h^2}{8 m L^2} \\ &= \frac{n^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (10 \times 10^{-10})^2} \\ E_n &= 0.603 \times 10^{-19} n^2 \text{ J} \\ \therefore E_n &= 0.38 n^2 \text{ eV} \end{aligned}$$

For $n = 1$, $E_1 = 0.38 \text{ eV}$

For $n = 2$, $E_2 = 1.52 \text{ eV}$

For $n = 3$, $E_3 = 3.42 \text{ eV}$

Example 4.13.5 An electron is trapped in a rigid box of width $4\text{\AA}$. Find its lowest energy level and momentum. Hence find energy of 3rd energy level.

Solution : $L = 4\text{\AA} = 4 \times 10^{-10} \text{ m}$

$$\begin{aligned} n = 1, \quad E_n &= \frac{n^2 h^2}{8 m L^2} = \frac{(1)^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (4 \times 10^{-10})^2} \\ &= 3.81 \times 10^{-19} \text{ J} \\ &= 2.381 \text{ eV} \end{aligned}$$

$$\begin{aligned} \therefore P_1 &= \sqrt{2 m E} \\ &= \sqrt{2 \times 9.1 \times 10^{-31} \times 3.81 \times 10^{-19} \text{ J}} \\ P_1 &= 8.327 \times 10^{-25} \text{ kg - m/sec.} \end{aligned}$$

$$\begin{aligned} \text{For } n = 3, \quad E_3 &= 3.81 \times 10^{-19} \times (3)^2 \\ &= 34.29 \times 10^{-19} \text{ J} \\ &= 21.43 \text{ eV.} \end{aligned}$$

Example 4.13.6 Compute energy difference between the ground state and first excited state for an electron in a one - dimensional rigid box of length 10^{-10} cm.

Solution :

$$L = 10^{-10} \text{ cm} = 10^{-12} \text{ m}$$

$$E_n = \frac{n^2 h^2}{8 mL^2}$$

$$E_1 = \frac{(1)^2 h^2}{8 mL^2}, \quad E_2 = \frac{(2)^2 h^2}{8 mL^2}$$

$$\begin{aligned} E_2 - E_1 &= \frac{4^2 h^2}{8 mL^2} - \frac{h^2}{8 mL^2} \\ &= \frac{3 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (10^{-12})^2} \\ &= 1.833 \times 10^{-13} \text{ J} \end{aligned}$$

$$E_2 - E_1 = 1.14 \times 10^6 \text{ eV}$$

Example 4.13.7 Proton and alpha particles are accelerated through same potential difference. Compute the ratio of their De-Broglie wavelength.

Solution :

$$M_\alpha = 4 M_p, e_\alpha = 2 e_p$$

$$\lambda = \frac{h}{\sqrt{2 m e V}}$$

For proton,

$$\lambda_p = \frac{h}{\sqrt{2 M_p e_p V}}$$

For α Particle,

$$\lambda_\alpha = \frac{h}{\sqrt{2 M_\alpha e_\alpha V}}$$

$$\lambda_\alpha = \frac{h}{\sqrt{4 \times 4 M_p e_p V}}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{h}{\sqrt{16 M_p e_p V}} \times \frac{\sqrt{2 M_p e_p V}}{h}$$

$$\frac{\lambda_\alpha}{\lambda_p} = \frac{1}{\sqrt{8}}$$

Example 4.13.8 An electron is confined to a box of length $10 \text{ \AA}$. Calculate the minimum uncertainty in its velocity.

Solution : $\Delta x = 10 \text{ \AA} = 10^{-8} \text{ cm}$

By Heisenberg's Uncertainty principle,

$$\Delta x \cdot \Delta p = \frac{h}{4\pi}$$

$$\therefore \Delta x \cdot (m \cdot \Delta V) = \frac{h}{4\pi}$$

$$\therefore \Delta V = \frac{h}{4\pi \times m \times \Delta x}$$

$$= \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 10 \times 10^{-8}}$$

$$\Delta V = 0.005800 \times 10^5 \text{ cm/s}$$

Example 4.13.9 An electron has a speed of 800 m/s with an accuracy of 0.007% . Calculate the uncertainty with which we can locate the position of the electron.

Solution : $V = 800 \text{ m/s}$, $\Delta V = 0.007 \% \text{ of } V = \frac{0.007}{100} \times 800$

$$\therefore \Delta x \cdot \Delta p = \frac{h}{4\pi}$$

$$\therefore \Delta x = \frac{h}{4\pi \cdot \Delta p} = \frac{h}{4\pi \cdot m \cdot \Delta V}$$

$$= \frac{6.63 \times 10^{-34} \times 100}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 0.007 \times 800}$$

$$\Delta x = 0.01035 \times 10^{-1} \text{ m}$$

Example 4.13.10 If a proton is moving with velocity 2% of the velocity of light, calculate the De-Broglie wavelength.

Solution : Given :

$$v = 2 \% \text{ of } c = 2 \times 10^{-2} \times 3 \times 10^8$$

$$= 6 \times 10^6 \text{ m/s}$$

$$\therefore \lambda = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{1.673 \times 10^{-27} \times 6 \times 10^6}$$

$$\lambda = 6.604 \times 10^{-14} \text{ m}$$

Example 4.13.11 An electron is trapped in an infinite potential well of width $1.75 \text{ \AA}$. Calculate energy difference between ground and first energy level.

Solution : Given : $L = 1.75 \text{ \AA} = 1.75 \times 10^{-10} \text{ m}$

$$E_n = \frac{n^2 h^2}{8 mL^2}$$

$$E_1 = \frac{(1)^2 h^2}{8 mL^2}, \quad E_2 = \frac{(2)^2 h^2}{8 mL^2}$$

$$E_2 - E_1 = \frac{3h^2}{8 mL^2} = \frac{3 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (1.75 \times 10^{-10})^2}$$

$$= 5.914 \times 10^{-18} \text{ J}$$

$$E_2 - E_1 = 36.96 \text{ eV}$$

Example 4.13.12 Calculate the energy difference between the ground state and first excited state of an electron in the rigid box of length $1 \text{ \AA}$.

Solution : Given : $L = 1 \text{ \AA} = 10^{-10} \text{ m}$.

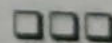
$$E_n = \frac{n^2 h^2}{8 mL^2}$$

$$E_1 = \frac{(1)^2 h^2}{8 mL^2}, \quad E_2 = \frac{(2)^2 h^2}{8 mL^2}$$

$$E_2 - E_1 = \frac{3h^2}{8 mL^2}$$

$$= \frac{3 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (10^{-10})^2}$$

$$= 114 \text{ eV}$$



2. An electron has a speed of 400 m/s with uncertainty of 0.01%. Find the accuracy in its position.

$$\text{Given : } v = 400 \frac{\text{m}}{\text{s}}, \quad \frac{\Delta v}{v} = \frac{0.01}{100}, \quad m = 9.1 \times 10^{-31} \text{ kg},$$

$$h = 6.63 \times 10^{-34}, \quad \Delta x = ?$$

$$p = m v = 9.1 \times 10^{-31} \times 400 = 3.64 \times 10^{-28} \frac{\text{kg.m}}{\text{s}}$$

$$\Delta p = m \Delta v = m v \frac{\Delta v}{v} = p \times \frac{\Delta v}{v} = 3.64 \times 10^{-28} \times \frac{0.01}{100} = 3.64 \times 10^{-32} \frac{\text{kg.m}}{\text{s}}$$

$$\text{According to HUP, } \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{\Delta p} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 3.64 \times 10^{-32}}$$

$$\Delta x = 1.449 \times 10^{-3} \text{ m}$$

3. An electron has a speed of 900 m/s with an accuracy of 0.001%. Calculate the uncertainty in the position of the electron.

$$\text{Given : } v = 900 \frac{\text{m}}{\text{s}}, \quad \frac{\Delta v}{v} = \frac{0.001}{100}, \quad m = 9.1 \times 10^{-31} \text{ kg},$$

$$h = 6.63 \times 10^{-34}, \quad \Delta x = ?$$

$$p = m v = 9.1 \times 10^{-31} \times 900 = 8.19 \times 10^{-28} \frac{\text{kg.m}}{\text{s}}$$

$$\Delta p = m \Delta v = m v \frac{\Delta v}{v} = p \times \frac{\Delta v}{v} = 8.19 \times 10^{-28} \times \frac{0.001}{100} = 8.19 \times 10^{-33} \frac{\text{kg.m}}{\text{s}}$$

$$\text{According to HUP, } \Delta x \cdot \Delta p \geq \frac{h}{4\pi}$$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{\Delta p} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 8.19 \times 10^{-33}}$$

$$\Delta x = 6.44 \times 10^{-3} \text{ m}$$

4. The speed of an electron is measured to within an uncertainty of 2×10^4 m/s.
What is the minimum space required by the electron to be confined to an atom?

$$\text{Given : } \Delta v = 2 \times 10^4 \frac{\text{m}}{\text{s}}, \quad m = 9.1 \times 10^{-31} \text{ kg},$$

$$h = 6.63 \times 10^{-34}, \quad \Delta x = ?$$

$$\text{According to HUP, } \Delta x \cdot \Delta p = \Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$$

$$\therefore \Delta x \geq \frac{h}{4\pi} \times \frac{1}{m \Delta v} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 9.1 \times 10^{-31} \times 2 \times 10^4}$$

$$\Delta x = 2.898 \times 10^{-9} \text{ m}$$

5. An electron confined in a box of length 10^{-8} m. Calculate minimum uncertainty in its velocity.

$$\text{Given : } \Delta x = 10^{-8} \text{ m}, \quad m = 9.1 \times 10^{-31} \text{ kg},$$

$$h = 6.63 \times 10^{-34}, \quad \Delta v = ?$$

$$\text{According to HUP, } \Delta x \cdot \Delta p = \Delta x \cdot m \Delta v \geq \frac{h}{4\pi}$$

$$\therefore \Delta v \geq \frac{h}{4\pi} \times \frac{1}{m \Delta x} = \frac{6.63 \times 10^{-34}}{4 \times 3.142 \times 9.1 \times 10^{-31} \times 10^{-8}}$$

$$\Delta v = 5797 \text{ m/s}$$

2. An electron is trapped in a one dimensional box of length 0.1 nm. Calculate the energy required to excite the electron from its ground state to the 4th excited state.

Given : $m = 9.1 \times 10^{-31} \text{ kg}$, $a = 0.1 \times 10^{-9} \text{ m}$

$h = 6.63 \times 10^{-34}$, $E_4 - E_0 = ?$

$$E = \frac{n^2 h^2}{8 m a^2}$$

For ground state, $n = 1$, $E = E_0$

$$\therefore E_0 = \frac{n^2 h^2}{8 m a^2} = \frac{1^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (0.1 \times 10^{-9})^2} = 6.04 \times 10^{-18} \text{ J}$$

For fourth excited state, $n = 5$, $E = E_4$

$$\therefore E_4 = \frac{n^2 h^2}{8 m a^2} = \frac{5^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (0.1 \times 10^{-9})^2} = 1.51 \times 10^{-16} \text{ J}$$

$$E_4 - E_0 = 1.51 \times 10^{-16} - 6.04 \times 10^{-18} = 1.45 \times 10^{-16} \text{ J}$$

$$= \frac{1.45 \times 10^{-16}}{1.6 \times 10^{-19}} = 906.25 \text{ eV}$$

3. An electron is bound by a potential which closely approaches an infinite square well of width $2.5 \times 10^{-10} \text{ m}$. Calculate the first lowest permissible energy for the electron.

Given : $m = 9.1 \times 10^{-31} \text{ kg}$, $a = 2.5 \times 10^{-10} \text{ m}$

$h = 6.63 \times 10^{-34}$, For lowest permissible Energy level, $E = E_1$ and $n = 2$

$$E = \frac{n^2 h^2}{8 m a^2}$$

For lowest permissible Energy level, $n = 2$, $E = E_1$

$$\therefore E_1 = \frac{n^2 h^2}{8 m a^2} = \frac{2^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2.5 \times 10^{-10})^2} = 3.864 \times 10^{-18} \text{ J}$$

1. An electron is bound in a one dimensional potential well of width $2 \text{ \AA}$ and of infinite height. Find its energy values in ground state and first two excited states.

$$\text{Given : } m = 9.1 \times 10^{-31} \text{ kg, } a = 2 \times 10^{-10} \text{ m}$$

$$h = 6.63 \times 10^{-34}, \quad E_0, E_1, E_2 = ?$$

$$E = \frac{n^2 h^2}{8 m a^2}$$

For ground state, $n = 1, E = E_0$

$$\therefore E_0 = \frac{n^2 h^2}{8 m a^2} = \frac{1^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} = 1.5 \times 10^{-18} \text{ J}$$

For first excited state, $n = 2, E = E_1$

$$\therefore E_1 = \frac{n^2 h^2}{8 m a^2} = \frac{2^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} = 6 \times 10^{-18} \text{ J}$$

For second excited state, $n = 3, E = E_2$

$$\therefore E_2 = \frac{n^2 h^2}{8 m a^2} = \frac{3^2 \times (6.63 \times 10^{-34})^2}{8 \times 9.1 \times 10^{-31} \times (2 \times 10^{-10})^2} = 1.35 \times 10^{-17} \text{ J}$$